Robust Repulsion for Growing Crowns in Linear Hypergraphs

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Abstract

Put $q = r-1$, $t = k-1$, and $D = tq+1$. For an edge e of a linear $C^r_{1,k}$-free r-uniform hypergraph, define
$$\delta_H(e) = \sum_{v\in e} \frac{1}{d_H(v)} - \frac{r}{D}.$$
The defect satisfies $\delta_H(e) \geq 0$. At equality, every vertex of e has degree D, and the petal trace at e is a disjoint union of t affine planes of order q. Equivalently, restoring the base line gives t projective planes of order q with common line e.

For growing crowns, the equality structure is stable in the following sense. If $q_j \to \infty$, $2 \leq t_j \leq q_j$, and e_j is an edge of a finite linear $C^{q_j+1}_{1,t_j+1}$-free hypergraph satisfying
$$\frac{t_j^2}{q_j} \to 0, \qquad t_j^3 \delta_{H_j}(e_j) \to 0,$$
then, for every fixed $0 < \theta < 1$,
$$\frac{|\{f \neq e_j : f \cap e_j \neq \varnothing,\ \delta_{H_j}(f) \geq \theta/t_j^2\}|}{(q_j+1)(t_jq_j)} \to 1.$$
Thus an edge close to equality is adjacent almost entirely to edges with defect of order at least t_j^{-2}. A uniform form gives absolute constants $Q_0, \varepsilon_0, c_0 > 0$ such that, whenever $q \geq Q_0t^2$ and $\delta_H(e) < \varepsilon_0/t^3$, at least $\frac{1}{2}(q+1)tq$ neighbors of e have defect at least $1/(100t^2)$. Consequently,
$$c^{\mathrm{lin}}_{q+1,t+1} \leq t - \frac{c_0}{t},$$
where $c^{\mathrm{lin}}_{r,k}$ denotes the asymptotic linear Turán coefficient for $C^r_{1,k}$.

1 Introduction

For $3 \leq k \leq r$, the r-uniform k-crown $C^r_{1,k}$ consists of a base edge e_0 and pairwise disjoint petals $e_1, \ldots, e_k$ satisfying
$$e_0 \cap e_i = \{v_i\} \qquad (i \in [k]),$$
where $v_1, \ldots, v_k$ are distinct points of e_0. Generalized crowns in linear hypergraphs were studied by Zhang, Broersma, and Wang [3]; Adak obtained the reciprocal-degree localization used below [1].

Set
$$q = r-1, \qquad t = k-1, \qquad D = tq+1, \qquad D^- = (t-1)q+1. \tag{1}$$

For an edge e, write

$$\Phi_H(e) = \sum_{v \in e} \frac{1}{d_H(v)}, \qquad \delta_H(e) = \Phi_H(e) - \frac{r}{D}.$$

The local bound is $\delta_H(e) \geq 0$. Equality forces every degree on e to equal D, and the petal trace decomposes into t affine planes of order q. A quantitative form of the localization argument is needed for stability; standard facts on affine and projective planes may be found in [2].

The stability problem is to control a second edge meeting an edge for which $\delta_H(e)$ is small. The useful error parameter is the number of uncovered point pairs in large subsets of the affine-plane structure forced by equality. Linearity makes these uncovered pairs a resource shared by external edges: two distinct external edges cannot use the same pair. A second low-defect edge, on the other hand, forces many external intersections. Comparing these upper and lower bounds yields the local repulsion theorem and, after double counting, a uniform gap in the asymptotic linear Turán coefficient.

2 Colored traces and the equality configuration

Fix an edge

$$e = \{v_1, \dots, v_r\}$$

of a linear r-uniform hypergraph H. For $i \in [r]$, define the i-th petal family

$$\mathcal{F}_i(e) = \{f \setminus \{v_i\} : f \in E(H) \setminus \{e\},\ v_i \in f\}.$$

Every member is a q-set outside e, and

$$x_i := |\mathcal{F}_i(e)| = d_H(v_i) - 1.$$

The index i is regarded as a color.

Lemma 2.1. *Each $\mathcal{F}_i(e)$ is a matching. If $i \neq j$, $A \in \mathcal{F}_i(e)$, and $B \in \mathcal{F}_j(e)$, then $|A \cap B| \leq 1$.*

Proof. Two distinct hyperedges through v_i already meet at v_i, so linearity makes their remaining q-sets disjoint. Hyperedges corresponding to different colors meet e at different points, and linearity permits at most one further intersection. □

A *rainbow matching* is a family of pairwise disjoint petals of distinct colors.

Proposition 2.2 (Crown–rainbow equivalence)**.** *The edge e is the base of a copy of $C^r_{1,k}$ if and only if $\mathcal{F}_1(e), \dots, \mathcal{F}_r(e)$ contain a rainbow matching of size k.*

Proof. Delete from each crown petal its point on e. This gives a rainbow matching. Conversely, adjoin the appropriate base point to every member of a rainbow matching. □

Let

$$\mathcal{M} = \{A_1, \dots, A_h\}$$

be a maximum rainbow matching, where A_j has color c_j, and put

$$U_{\mathcal{M}} = A_1 \cup \cdots \cup A_h.$$

Every petal of a color not used by $\mathcal{M}$ meets $U_{\mathcal{M}}$, or the matching could be enlarged. For such a petal A, define

$$T_{\mathcal{M}}(A) = \{j \in [h] : A \cap A_j \neq \varnothing\}$$

and

$$J_{\mathcal{M}} = \sum_{\substack{A \text{ of a color} \\ \text{unused by } \mathcal{M}}} \big(|T_{\mathcal{M}}(A)| - 1\big) \geq 0.$$

Proposition 2.3 (Matching-blocker inequality). *Let $C(\mathcal{M}) = \{c_1, \dots, c_h\}$. Then*

$$\sum_{i \notin C(\mathcal{M})} x_i + J_{\mathcal{M}} \leq \sum_{w \in U_{\mathcal{M}}} \min\{r - h, d_H(w) - 1\}.$$

Proof. The left side is the total number of incidences between unused-color petals and $U_{\mathcal{M}}$. Indeed, the selected petals are disjoint and a petal meets each of them in at most one point. At a fixed $w \in U_{\mathcal{M}}$, at most one petal from each unused color contains w, giving at most $r - h$ incidences. Exactly one selected hyperedge contains w, so at most $d_H(w) - 1$ other hyperedges can contribute. Summing the smaller bound proves the proposition. □

2.1 The equality case for colored traces

Consider an abstract collection

$$\mathcal{F}_1, \dots, \mathcal{F}_r$$

of matchings of q-sets. Sets of different colors meet in at most one point. Assume

$$|\mathcal{F}_i| = tq \qquad (i \in [r])$$

and that there is no rainbow matching of size $t + 1$.

Lemma 2.4 (Blocking at equality). *Let $\mathcal{M}$ be a rainbow matching of size t. For every unused color, its tq sets meet $U_{\mathcal{M}}$ in singletons, and those singletons partition $U_{\mathcal{M}}$.*

Proof. Every unused-color set meets $U_{\mathcal{M}}$, or it augments $\mathcal{M}$. The tq sets of one color are disjoint, and their nonempty intersections with the tq-point set $U_{\mathcal{M}}$ are disjoint. Hence all intersections are singletons and cover the union. □

Lemma 2.5 (Extendability). *Every rainbow matching of size at most t extends to one of size t. If a color is unused by the initial matching, the extension can be chosen to avoid that color.*

Proof. Suppose the current matching has size $h < t$, and let j be a color that is required to remain unused. At most $h + 1 \leq t$ colors are used or forbidden. Since $r = q + 1 \geq t + 1$, some color i is neither. Each selected q-set meets at most q members of the matching $\mathcal{F}_i$, so the h selected sets block at most $hq < tq = |\mathcal{F}_i|$ members of $\mathcal{F}_i$. Add an unblocked member and repeat. □

Join two sets of different colors when they intersect.

Lemma 2.6 (Exact pair degree). *Every set meets exactly q sets of every other color.*

Proof. A q-set meets at most q members of a matching. Fix $A \in \mathcal{F}_i$ and another color j. Extend $\{A\}$ to a rainbow t-matching avoiding color j, using Lemma 2.5. By Lemma 2.4, the color-j sets partition the matching union by singleton intersections. Exactly one passes through each point of A. □

Lemma 2.7 (Transitivity of intersections). *Let A, B, C have three distinct colors. If A meets both B and C, then B meets C.*

Proof. If B and C were disjoint, extend $\{B, C\}$ to a rainbow t-matching avoiding the color of A. The set A would meet the matching union at least twice, contrary to Lemma 2.4. $\square$

Lemma 2.8 (Pairwise block decomposition)**.** *For every pair of colors, the bipartite intersection graph between them is a disjoint union of t copies of $K_{q,q}$.*

Proof. By Lemma 2.6, the intersection graph is q-regular with tq vertices in each part. Suppose A, A' in the first part have a common neighbor B. Choose a third color and a set C of that color meeting A. For any neighbor B' of A, repeated applications of Lemma 2.7 give

$$A \sim B,\ A \sim C \Rightarrow B \sim C, \qquad A \sim B',\ A \sim C \Rightarrow B' \sim C,$$
$$B \sim A',\ B \sim C \Rightarrow A' \sim C, \qquad C \sim A',\ C \sim B' \Rightarrow A' \sim B'.$$

Thus $N(A) \subseteq N(A')$, and equality follows from regularity. Along every even path, all vertices in the same part have the same neighborhood. Every connected component is therefore complete bipartite. Regularity makes it $K_{q,q}$, and there are t components. $\square$

Lemma 2.9 (Common labeling)**.** *There are partitions*

$$\mathcal{F}_i = \mathcal{F}_i^1 \dot\cup \cdots \dot\cup \mathcal{F}_i^t, \qquad |\mathcal{F}_i^b| = q,$$

such that two sets of different colors intersect if and only if they have the same label.

Proof. Fix color 1. For each $j \neq 1$, Lemma 2.8 partitions $\mathcal{F}_1$ into common-neighborhood classes of size q. It remains to verify that this partition is independent of j.

Suppose $A, A' \in \mathcal{F}_1$ have a common color-j neighbor B. Let $\ell \notin \{1, j\}$ and choose $C \in \mathcal{F}_\ell$ meeting A. Transitivity first gives $B \sim C$, and then $A' \sim C$. Hence every color-ℓ neighbor of A is a neighbor of A'. Both have q such neighbors, so the neighborhoods are equal. The same is true for every nonreference color. The partitions of $\mathcal{F}_1$ therefore coincide.

Label the common classes $1, \ldots, t$. A set of another color receives the label of its neighborhood in $\mathcal{F}_1$. Every label class has size q. If two nonreference sets have the same label, a member of the corresponding reference class meets both, so transitivity makes them intersect. Conversely, if they intersect, transitivity with a reference neighbor shows that their reference neighborhoods coincide. $\square$

Theorem 2.10 (Colored affine-plane classification)**.** *Under the hypotheses of this subsection, the colored system is the disjoint union of t affine planes of order q. More precisely, for every $b \in [t]$ there is a q^2-point set Ω_b such that*

$$\bigcup_{i=1}^{r} \mathcal{F}_i^b$$

is the block set of an affine plane of order q, with the $\mathcal{F}_i^b$ as its $q+1$ parallel classes. The sets $\Omega_1, \ldots, \Omega_t$ are pairwise disjoint.

Proof. Fix b and a color i. The q members of $\mathcal{F}_i^b$ are disjoint q-sets, so their union Ω_b has size q^2. If $j \neq i$ and $B \in \mathcal{F}_j^b$, then B meets every member of $\mathcal{F}_i^b$. These q intersections are distinct and exhaust B, so $B \subseteq \Omega_b$. Thus every $\mathcal{F}_j^b$ is a parallel class partitioning the same point set.

There are $q(q+1)$ blocks. No pair of points occurs twice, and the number of covered pairs is

$$q(q+1)\binom{q}{2} = \binom{q^2}{2}.$$

Every pair is therefore covered once. The blocks form an $S(2, q, q^2)$, equivalently an affine plane of order q. Different labels are disjoint by Lemma 2.9. $\square$

3 Reciprocal localization and equality

The following reciprocal-degree localization is used in the stability argument and is included to keep the crown analysis self-contained.

Lemma 3.1 (Greedy crown lemma)**.** *Let G be a linear r-uniform hypergraph. Suppose that an edge e contains distinct vertices $v_1, \ldots, v_k$ satisfying*

$$d_G(v_i) \geq (i-1)(r-1) + 2 \qquad (i \in [k]).$$

Then e is the base of a copy of $C^r_{1,k}$.

Proof. Choose the petals in the order $v_1, \ldots, v_k$. Suppose that pairwise disjoint petals $f_1, \ldots, f_{i-1}$ have been chosen, with $f_j \cap e = \{v_j\}$. There are at least $(i-1)(r-1)+1$ candidate edges other than e through v_i. For fixed $j < i$, the edge f_j blocks at most $r-1$ candidates: each candidate meeting f_j uses one of the $r-1$ points of $f_j \setminus \{v_j\}$, and linearity permits at most one candidate through v_i and any prescribed such point. The previously chosen petals block at most $(i-1)(r-1)$ candidates. A candidate remains, and induction gives the required crown. $\square$

Lemma 3.2 (Degree witness)**.** *Let H be linear and $C^r_{1,k}$-free. If $e = \{u_1, \ldots, u_r\}$ is ordered so that $d_H(u_1) \geq \cdots \geq d_H(u_r)$, then some $i \in [k]$ satisfies*

$$d_H(u_i) \leq (k-i)(r-1) + 1.$$

Proof. Otherwise set $v_j = u_{k-j+1}$ for $j \in [k]$. The reversed vertices satisfy the degree condition in Lemma 3.1, a contradiction. $\square$

Proposition 3.3 (Edgewise localization)**.** *Every edge e of a linear $C^r_{1,k}$-free r-graph satisfies*

$$\Phi_H(e) \geq \frac{r}{D}.$$

Equality holds if and only if every vertex of e has degree D.

Proof. Order $e = \{u_1, \ldots, u_r\}$ by decreasing degree, take the witness i in Lemma 3.2, and put $B_i = (k-i)(r-1)+1$. Then $d_H(u_j) \leq B_i$ for $j \geq i$, whence

$$\Phi_H(e) \geq \frac{r-i+1}{B_i}.$$

A direct subtraction gives

$$\frac{r-i+1}{B_i} - \frac{r}{D} = \frac{(i-1)\{r(r-k)+k-2\}}{B_i D}.$$

The numerator in braces is nonnegative because $r(r-k)+k-2 \geq r-2 > 0$. This proves the required inequality.

Equality forces $i = 1$ in the displayed gap identity. Then $d_H(u_1) \leq D$, hence every degree on e is at most D, and $\sum_{u \in e} 1/d_H(u) = r/D$ forces all of them to equal D. The converse is immediate. $\square$

For an edge e, recall

$$\delta_H(e) = \Phi_H(e) - \frac{r}{D}, \qquad L(e) = \sum_{v \in e} (D - d_H(v)).$$

The summands defining $L(e)$ will only be used after nonnegativity has been established. Put

$$A_{q,t} = q^2 - (t-1)q - 1, \qquad G_{q,t} = \frac{A_{q,t}}{DD^-}.$$

Since $q \geq t$, one has $A_{q,t} > 0$.

Proposition 3.4 (Degree stability)**.** *If $\delta_H(e) < G_{q,t}$, then every vertex of e has degree at most D, and*

$$L(e) \leq D^2 \delta_H(e).$$

Proof. For a witness of index $i \geq 2$, the displayed gap identity, with $r = q+1$ and $k = t+1$, gives

$$\delta_H(e) \geq \frac{(i-1)A_{q,t}}{((t+1-i)q+1)D}.$$

The ratio $(i-1)/((t+1-i)q+1)$ is strictly increasing, so its minimum for $i \geq 2$ occurs at $i = 2$ and equals $1/D^-$. Thus $\delta_H(e) < G_{q,t}$ forces a witness of index one. As in the equality proof, all degrees on e are then at most D. For $d \leq D$,

$$\frac{1}{d} - \frac{1}{D} = \frac{D-d}{Dd} \geq \frac{D-d}{D^2}.$$

Summing over e proves the claim. □

Theorem 3.5 (Equality structure)**.** *Let e be an edge of a linear $C^{q+1}_{1,t+1}$-free hypergraph. The following are equivalent:*

1. $\delta_H(e) = 0$;
2. *every point of e has degree D;*
3. *the petal trace at e is a disjoint union of t affine planes of order q, with the colors as parallel classes.*

For each affine packet, adjoining the base point v_i to every line of color i, and adjoining the base edge e, produces a projective plane of order q. The resulting t projective planes have common line e and are otherwise disjoint.

Proof. The equivalence of the first two statements is Proposition 3.3. If every point of e has degree D, then every color has $D - 1 = tq$ petals. By Proposition 2.2, crown-freeness excludes a rainbow matching of size $t+1$, and Theorem 2.10 gives the affine-plane decomposition. Conversely, the decomposition supplies tq petals of every color, so every point of e has degree D.

Fix one affine packet. Add the $q+1$ points of e as points at infinity, assign v_i to the parallel class of color i, and take e as the line at infinity. Lines in one affine class meet at their assigned point, lines in distinct classes retain their unique affine intersection, and every affine point lies on one line of each class. This is a projective plane of order q. The affine packets are disjoint, so the projective planes intersect exactly in e. □

4 Approximate blocking near equality

Assume throughout this section that $e = \{v_1, \ldots, v_r\}$ satisfies the hypothesis of Proposition 3.4. Write

$$\Delta_i = D - d_H(v_i), \qquad x_i = |\mathcal{F}_i(e)| = tq - \Delta_i, \qquad \sum_{i=1}^{r} \Delta_i = L(e).$$

Thus all Δ_i are nonnegative integers.

Proposition 4.1 (Approximate blocking)**.** *Suppose*

$$L(e) < q(q - t + 2). \tag{2}$$

Then the trace at e has a maximum rainbow matching $\mathcal{M} = \{A_1, \dots, A_t\}$ of size exactly t. Put $U = A_1 \cup \cdots \cup A_t$. For every color i unused by $\mathcal{M}$:

1. *at most Δ_i members of $\mathcal{F}_i(e)$ meet at least two branches A_b;*
2. *at most Δ_i points of U are uncovered by color i;*
3. *for every $b \in [t]$, at least $q - 2\Delta_i$ color-i rows have trace on U equal to a singleton in A_b;*
4. *singleton rows of two distinct unused colors with the same branch label intersect;*
5. *if B is a singleton row of label b and $j \neq i$ is another unused color, then B meets at most $2\Delta_j$ color-j rows not having singleton label b.*

Proof. Let $h \leq t$ be the size of a maximum rainbow matching and let $U_{\mathcal{M}}$ be its union. The unused colors contain at least

$$(r - h)tq - L(e)$$

rows. Proposition 2.3, with $|U_{\mathcal{M}}| = hq$, bounds this number above by $hq(r - h)$. If $h < t$, then

$$L(e) \geq (t - h)q(r - h) \geq q(q - t + 2),$$

contrary to (2). Hence $h = t$.

Fix an unused color i. Maximality makes every one of its $tq - \Delta_i$ rows meet U. Since the color is a matching, its row traces on the tq-point set U are disjoint. Therefore

$$\sum_{B \in \mathcal{F}_i(e)} (|B \cap U| - 1) \leq \Delta_i.$$

This proves part 1. The nonempty disjoint traces cover at least $tq - \Delta_i$ points, proving part 2. On a fixed branch A_b, at most Δ_i points are uncovered and at most Δ_i more belong to nonsingleton traces. Every remaining point lies in a distinct singleton row of label b, proving part 3.

For part 4, suppose singleton rows B, C of distinct unused colors and common label b were disjoint. Then

$$\{B, C\} \cup \{A_a : a \neq b\}$$

would be a rainbow matching of size $t + 1$, a contradiction. Finally, part 3 supplies at least $q - 2\Delta_j$ color-j singleton rows of label b. Part 4 makes all of them meet B, at distinct points because color j is a matching. At most $2\Delta_j$ points of B remain available to rows of color j with another trace type. This proves part 5. $\square$

5 Uncovered pairs

Let $\mathcal{R}$ be a linear family of sets, let Ω be a point set, and write $s_B = |B \cap \Omega|$. Let M_Ω be the incidence matrix of the restricted rows $B \cap \Omega$, and let $d_\Omega(x)$ be the corresponding column degree. Define

$$\mathcal{U}(\Omega) = \binom{|\Omega|}{2} - \sum_{B \in \mathcal{R}} \binom{s_B}{2}.$$

Proposition 5.1 (Pair-defect identity)**.** *One has* $\mathcal{U}(\Omega) \geq 0$ *and*

$$\boxed{\mathcal{U}(\Omega) = \binom{|\Omega|}{2} - \frac{1}{2}\left\| M_\Omega^\top M_\Omega - \operatorname{diag}(d_\Omega(x) : x \in \Omega)\right\|_F^2.}$$

Suppose that the rows in $\mathcal{R}$ *arise from hyperedges in a linear hypergraph. If every hyperedge* g *in a family* $\mathcal{E}$ *is distinct from those row-edges, then*

$$\#\{g \in \mathcal{E} : |g \cap \Omega| \geq h\} \leq \frac{\mathcal{U}(\Omega)}{\binom{h}{2}} \qquad (h \geq 2),$$

provided every pair of $g \cap \Omega$ *is required to be uncovered by* $\mathcal{R}$*. This requirement holds, in particular, when the row-edges all meet a fixed edge* e *and the members of* $\mathcal{E}$ *are disjoint from* e*.*

Proof. Linearity makes the two-point subsets contributed by distinct restricted rows disjoint, proving nonnegativity. Every off-diagonal entry of $M_\Omega^\top M_\Omega$ is zero or one, and it equals one precisely on a covered ordered pair. Its diagonal is $d_\Omega(x)$, so the squared Frobenius norm in the displayed identity is twice the number of covered unordered pairs.

If all pairs in $g \cap \Omega$ are uncovered, then g consumes at least $\binom{h}{2}$ uncovered pairs. Distinct hyperedges consume disjoint pairs, since otherwise they share two points. This proves the claimed bound. In the stated special case, a pair contained both in g and in a row-edge would make those two hyperedges intersect twice. □

6 Packet construction and uncovered pairs

Assume the hypotheses of Proposition 4.1, and fix a maximum rainbow matching $\mathcal{M} = \{A_1, \dots, A_t\}$. Let J be the set of unused colors, so

$$p_0 := |J| = q + 1 - t.$$

For $i \in J$ and $b \in [t]$, let $\mathcal{F}_i^b$ be the rows of color i whose trace on $A_1 \cup \cdots \cup A_t$ is a singleton in A_b. Put

$$\xi = \frac{L(e)}{q^2}.$$

Theorem 6.1 (Packet construction and uncovered pairs)**.** *Suppose* $p_0 \geq 2$*. Choose* $p, c \in J$ *with the two smallest values of* Δ_i*, and define*

$$w = 2(\Delta_p + \Delta_c), \qquad \rho = \frac{w}{q}.$$

Then

$$\rho \leq \frac{4L(e)}{q(q+1-t)}. \tag{3}$$

Assume $\rho < 1$*, and for* $b \in [t]$ *set*

$$\Omega_b = \{R \cap C : R \in \mathcal{F}_p^b,\ C \in \mathcal{F}_c^b\}.$$

The sets $\Omega_1, \dots, \Omega_t$ *are pairwise disjoint and satisfy:*

1.

$$|\Omega_b| \geq (1 - \rho)q^2;$$

2. every unused-color singleton row of label b contains at least $(1-\rho)q$ points of Ω_b;

3. if

$$\mathcal{R}_b = \bigcup_{i\in J} \mathcal{F}_i^b,$$

then

$$|\mathcal{R}_b| \geq q^2 - (t-1)q - 2L(e);$$

4. with $s_B = |B \cap \Omega_b|$, define

$$\mathcal{U}_b = \binom{|\Omega_b|}{2} - \sum_{B\in\mathcal{R}_b} \binom{s_B}{2}.$$

Then

$$0 \leq \mathcal{U}_b \leq \frac{\eta_* q^4}{2}, \qquad \eta_* = \frac{t}{q} + 2\xi + 2\rho;$$

5. let $\mathcal{E}$ be any family of hyperedges disjoint from e, and put $h_b(g) = |g \cap \Omega_b|$. If $\beta q > t$, then

$$\left|\left\{ g \in \mathcal{E} : \sum_{b=1}^{t} h_b(g) \geq \beta q \right\}\right| \leq \frac{t^2 \eta_*}{\beta(\beta - t/q)}\, q^2.$$

Moreover,

$$\eta_* \leq \frac{t}{q} + 2\xi + \frac{8\xi}{1-(t-1)/q}. \tag{4}$$

Proof. The average of the two smallest nonnegative values is no larger than the average of all values, so

$$\Delta_p + \Delta_c \leq \frac{2}{p_0} \sum_{i\in J} \Delta_i \leq \frac{2L(e)}{q+1-t}.$$

This proves (3). Proposition 4.1 gives

$$|\mathcal{F}_i^b| \geq q - 2\Delta_i \qquad (i \in J,\ b \in [t]). \tag{5}$$

Every p-row and c-row with the same label intersects, and the intersection map is injective. Hence

$$|\Omega_b| = |\mathcal{F}_p^b|\,|\mathcal{F}_c^b| \geq (q-2\Delta_p)(q-2\Delta_c) \geq q^2 - wq.$$

Packet disjointness follows from the color-p matching.

Let $B \in \mathcal{F}_i^b$. If $i \notin \{p, c\}$, then B meets every retained p-row and every retained c-row. Inclusion–exclusion inside the q-set B gives

$$|B \cap \Omega_b| \geq |\mathcal{F}_p^b| + |\mathcal{F}_c^b| - q \geq q - w.$$

The cases $i = p, c$ follow by intersecting with the opposite defining class. This proves part 2. Summing (5) over $i \in J$ gives

$$|\mathcal{R}_b| \geq (q+1-t)q - 2\sum_{i\in J} \Delta_i \geq q^2 - (t-1)q - 2L(e),$$

which is part 3.

Part 4 uses Proposition 5.1. Since $|\Omega_b| \le q^2$ and every row in $\mathcal{R}_b$ has at least $q-w$ points in Ω_b,

$$\mathcal{U}_b \le \binom{q^2}{2} - \left(q^2 - (t-1)q - 2L(e)\right)\binom{q-w}{2}.$$

Put

$$x = \frac{t-1}{q} + 2\xi, \qquad y = 1 - \left((1-\rho)^2 - \frac{1}{q}\right).$$

If $\eta_* \ge 1$, the claimed upper bound follows from $\mathcal{U}_b \le \binom{q^2}{2}$. Otherwise $x, y \in [0,1]$, $y \le 2\rho + 1/q$, and $x + y \le \eta_*$. The two factors subtracted above are at least $q^2(1-x)$ and $q^2(1-y)/2$, respectively. Since $(1-x)(1-y) \ge 1-x-y$, part 4 follows.

For part 5, an external edge g consumes

$$\sum_{b=1}^{t} \binom{h_b(g)}{2}$$

uncovered within-packet pairs, and distinct external edges consume disjoint pairs. If

$$H_g = \sum_{b=1}^{t} h_b(g) \ge \beta q,$$

then Cauchy–Schwarz gives

$$\sum_{b=1}^{t} \binom{h_b(g)}{2} = \frac{1}{2}\left(\sum_b h_b(g)^2 - H_g\right) \ge \frac{1}{2}\left(\frac{H_g^2}{t} - H_g\right) \ge \frac{\beta q(\beta q - t)}{2t}.$$

Since

$$\sum_{b=1}^{t} \mathcal{U}_b \le \frac{t\eta_* q^4}{2},$$

division gives the stated bound.

Finally, (3) and $L(e) = \xi q^2$ give

$$2\rho \le \frac{8\xi}{1-(t-1)/q},$$

which proves (4). □

7 Prescribed-petal intersections

The following bound links the reciprocal defect at one edge to the uncovered-pair estimate associated with another. The first petal in the resulting rainbow matching is prescribed.

Theorem 7.1 (Prescribed-petal intersection bound)**.** *Let f be an edge of a linear $C_{1,t+1}^{q+1}$-free hypergraph, suppose $d_H(z) \le D$ for every $z \in f$, and put*

$$\Delta_z = D - d_H(z), \qquad L(f) = \sum_{z \in f} \Delta_z.$$

Fix $x \in f$, and assume

$$L(f) < q(q-t+1).$$

If $g \neq f$ meets f at a point different from x, then g meets at least $q - \Delta_x$ distinct edges in

$$\mathcal{E}_x(f) = \{h \neq f : x \in h\}.$$

The corresponding intersection points on g are distinct.

Proof. Suppose $g \cap f = \{y\}$, where $y \neq x$. Seed a rainbow matching in the petal trace at f with $g \setminus \{y\}$, and extend it maximally using only the q colors in $f \setminus \{x\}$. Let its size be $h \leq t$, and let U be the union of its h disjoint q-set traces.

If $h < t$, there are $q - h$ unused permitted colors. They contain at least

$$(q-h)tq - L(f)$$

petals. Every such petal meets U, by maximality. At each point of U, at most one petal of each unused color occurs. Therefore

$$(q-h)tq - L(f) \leq |U|(q-h) = hq(q-h),$$

and hence

$$L(f) \geq q(q-h)(t-h).$$

For $h \leq t-1$, the right side is at least $q(q-t+1)$: the function $(q-h)(t-h)$ decreases with h on this range and has value $q-t+1$ at $h = t-1$. This contradicts the assumed bound on $L(f)$, so $h = t$.

Write the selected actual hyperedges as $g = g_1, g_2, \ldots, g_t$. There are

$$|\mathcal{E}_x(f)| = d_H(x) - 1 = tq - \Delta_x$$

edges in the opposite star. Every one meets at least one g_j, for otherwise that edge together with $g_1, \ldots, g_t$ would be a $(t+1)$-crown with base f. Each g_j meets at most q members of $\mathcal{E}_x(f)$: its point on f cannot lie on an x-edge, and each of its other q points lies on at most one x-edge by linearity. The last $t-1$ selected edges therefore meet at most $(t-1)q$ members of the opposite star. The prescribed edge $g_1 = g$ meets at least

$$tq - \Delta_x - (t-1)q = q - \Delta_x$$

of them. Two such opposite-star edges cannot meet g at the same point, since they already share x. $\square$

8 Comparing external demand with uncovered pairs

Fix $0 < a < 1$. An unused color i is *good* if $\Delta_i \leq aq$. For a good color i, let X_i contain every point of every nonsingleton color-i trace row and, from each singleton row of label b, every point outside Ω_b. A neighbor f of e is *retained* if its color is unused and good and its trace row has a singleton branch label.

Lemma 8.1 (Spill bound). *For every good unused color i,*

$$|X_i| \leq (a + t\rho)q^2.$$

Let f be a retained neighbor of e, let $x = e \cap f$, and suppose $d_H(z) \leq D$ for every $z \in f \setminus \{x\}$. Let $\mathcal{E}_f$ be the set of edges meeting f away from x and disjoint from e. For every point set X outside e, the number of incidences

$$\{(g, z) : g \in \mathcal{E}_f,\ z \in g \cap X\}$$

is at most

$$q|X| + (t-1)q^2.$$

Proof. At most Δ_i rows of color i are nonsingleton, so they contribute at most $q\Delta_i \le aq^2$ points. By Theorem 6.1(2), every singleton row has at most ρq points outside its packet, and color i has at most tq rows. This proves the first assertion.

If $z \notin f$, at most one edge through z can correspond to each of the q possible intersection points in $f \setminus \{x\}$, so its multiplicity in $\mathcal{E}_f$ is at most q. If $z \in f \setminus \{x\}$, its multiplicity is at most $d_H(z) - 1 \le D - 1$. Since $X \cap f$ has at most q points, the incidence count is at most

$$q|X \setminus f| + (D-1)|X \cap f| \le q|X| + (D-1-q)q = q|X| + (t-1)q^2.$$

□

Theorem 8.2 (Uncovered-pair criterion)**.** *Assume Theorem 6.1. Choose*

$$0 < a < 1, \qquad \frac{t}{q} < \beta < 1 - a,$$

and $b > 0$ satisfying

$$b < G_{q,t}, \qquad D^2 b < q(q-t+1).$$

If

$$(t-1) - \frac{D^2 b}{q^2} > \frac{a + t\rho + (t-1)/q}{1-a-\beta} + \frac{t^2\eta_*}{\beta(\beta - t/q)}, \tag{6}$$

then every retained neighbor f of e satisfies $\delta_H(f) \ge b$.

Proof. Suppose that a retained neighbor f satisfies $\delta_H(f) < b$, and put $x = e \cap f = v_i$. Since $b < G_{q,t}$, Proposition 3.4 applied to f gives degree at most D at every point of f and

$$L(f) \le D^2 \delta_H(f) < D^2 b < q(q-t+1).$$

For each $y \in f \setminus \{x\}$, at most q edges through y, other than f, meet e. Hence

$$|\mathcal{E}_f| \ge \sum_{y \in f \setminus \{x\}} (d_H(y) - 1 - q) = (t-1)q^2 - L(f) + \Delta_i > \left((t-1) - \frac{D^2 b}{q^2}\right) q^2.$$

Take $g \in \mathcal{E}_f$. Theorem 7.1, applied to f with the prescribed edge g, gives at least $q - \Delta_i \ge (1-a)q$ distinct intersections with color-i trace rows at e. Every such intersection lies in

$$\Omega := \Omega_1 \cup \cdots \cup \Omega_t$$

or in X_i. Call g *light* if $|g \cap \Omega| < \beta q$. A light edge contains more than $(1-a-\beta)q$ points of X_i. Lemma 8.1 therefore gives

$$|\mathcal{E}_f^{\mathrm{light}}| \le \frac{a + t\rho + (t-1)/q}{1-a-\beta}\, q^2.$$

Every remaining edge contains at least βq packet points. Theorem 6.1(5) bounds their number by

$$\frac{t^2\eta_*}{\beta(\beta - t/q)}\, q^2.$$

Inequality (6) contradicts the lower bound for $|\mathcal{E}_f|$. □

9 Growing-crown repulsion

Theorem 9.1 (Asymptotic local repulsion). *Let $q_j \to \infty$, let $2 \le t_j \le q_j$, and put*

$$r_j = q_j + 1, \qquad D_j = t_j q_j + 1.$$

Let e_j be an edge of a finite linear $C^{r_j}_{1,t_j+1}$-free r_j-graph H_j. Assume

$$\frac{t_j^2}{q_j} \longrightarrow 0, \qquad t_j^3 \delta_{H_j}(e_j) \longrightarrow 0.$$

Then, for every fixed $0 < \theta < 1$,

$$\frac{\left|\{f \ne e_j : f \cap e_j \ne \varnothing,\ \delta_{H_j}(f) \ge \theta/t_j^2\}\right|}{r_j(D_j - 1)} \longrightarrow 1.$$

Proof. Suppress the index j. Under $t^2/q \to 0$,

$$t^2 G_{q,t} = \frac{t^2(q^2 - (t-1)q - 1)}{(tq+1)((t-1)q+1)} \ge 1 - o(1).$$

The second hypothesis implies $\delta_H(e) < G_{q,t}$ for all large j. Proposition 3.4 gives

$$\xi = \frac{L(e)}{q^2} \le \left(\frac{D}{q}\right)^2 \delta_H(e), \qquad t\xi \longrightarrow 0.$$

The condition of Proposition 4.1 holds eventually, $p_0 = q + 1 - t \ge 2$, and the packet construction is available. By (3) and (4),

$$t\rho \longrightarrow 0, \qquad \eta_* = \frac{t}{q} + O(\xi + \rho), \qquad t\eta_* \longrightarrow 0.$$

Set

$$\beta = \frac{1}{2}, \qquad b = \frac{\theta}{t^2},$$

and choose a fixed

$$0 < a < \min\left\{\frac{1}{8}, \frac{1-\theta}{16}\right\}.$$

The two threshold conditions in Theorem 8.2 hold for all large j: one has $b < G_{q,t}$, while

$$\frac{D^2 b}{q^2} = \theta\left(1 + \frac{1}{tq}\right)^2 < 1 - \frac{t-1}{q}.$$

For (6), the left side is $t - 1 - \theta - o(1)$. The first term on the right is $a/(1/2 - a) + o(1)$. The second term is $o(t)$, because $t\eta_* \to 0$. If t is bounded, it is $o(1)$; if $t \to \infty$, divide the inequality by t. The choice of a then makes (6) valid for all large j. Thus every retained neighbor has defect at least θ/t^2.

It remains to count retained neighbors. The t colors used by the blocking matching contain at most $t^2 q$ rows. The bad unused colors contain at most $(t/a)L(e)$ rows, and the nonsingleton rows in the good unused colors contribute at most $L(e)$. Hence at most

$$t^2 q + \left(\frac{t}{a} + 1\right) L(e)$$

neighboring edges are discarded. The actual number of neighbors of e is

$$\sum_{v\in e}(d_H(v)-1)=r(D-1)-L(e).$$

Dividing the total shortage by $r(D-1)=tq(q+1)$ gives

$$O\left(\frac{t}{q}+\frac{\xi}{a}+\frac{\xi}{t}\right)=o(1).$$

The claimed limit follows. $\square$

Corollary 9.2 (Uniform local repulsion)**.** *There are absolute constants $Q_0<\infty$ and $\varepsilon_0>0$ such that the following holds. If*

$$q\geq Q_0t^2$$

and e is an edge of a finite linear $C_{1,t+1}^{q+1}$-free hypergraph with

$$\delta_H(e)<\frac{\varepsilon_0}{t^3},$$

then at least

$$\frac{1}{2}r(D-1)$$

neighbors f of e satisfy

$$\delta_H(f)\geq\frac{1}{100t^2}.$$

Proof. Set

$$a=\frac{1}{100},\qquad\beta=\frac{1}{2},\qquad b=\frac{1}{100t^2}.$$

Choose $\varepsilon_0>0$ so that $\varepsilon_0\leq 1/100$ and $132\varepsilon_0<1/100$. Then choose Q_0 so large that all inequalities used below, including the final retention estimate, hold whenever $q\geq Q_0t^2$. Assume $\delta_H(e)<\varepsilon_0/t^3$. Then $\delta_H(e)<G_{q,t}$, so Proposition 3.4 gives

$$\xi\leq\left(\frac{D}{q}\right)^2\frac{\varepsilon_0}{t^3}\leq\frac{2\varepsilon_0}{t}.$$

Equation (3) also gives

$$\rho\leq\frac{9\varepsilon_0}{t},$$

and therefore

$$t\eta_*=\frac{t^2}{q}+2t\xi+2t\rho\leq Q_0^{-1}+22\varepsilon_0.$$

The same choice of Q_0 ensures $b<G_{q,t}$ and $D^2b<q(q-t+1)$.

It remains to verify (6). Its first term on the right is at most

$$\frac{1/100+9\varepsilon_0+1/(2Q_0)}{49/100}.$$

Moreover $t/q\leq 1/(2Q_0)$, and hence its second term is at most

$$6t\big(Q_0^{-1}+22\varepsilon_0\big).$$

The left side is at least

$$t-1-\frac{1}{100}\left(1+\frac{1}{tq}\right)^2.$$

The choice of ε_0 makes the terms independent of Q_0 sufficiently small, and the choice of Q_0 makes the remaining Q_0^{-1}-terms small enough that the right side is below the displayed left side for every $t \geq 2$. Hence Theorem 8.2 applies uniformly.

The number of missing or nonretained neighbors is at most

$$L(e)+t^2q+\left(\frac{t}{a}+1\right)L(e).$$

Dividing by $r(D-1)=tq(q+1)$ gives at most

$$\frac{t}{q+1}+\left(\frac{1}{a}+\frac{2}{t}\right)\xi.$$

By the same choices of ε_0 and Q_0, this quantity is less than $1/2$. Thus at least half of the neighbors are retained, and Theorem 8.2 gives $\delta_H(f) \geq 1/(100t^2)$ on all of them. □

10 A uniform Turán gap

For a finite hypergraph H, put

$$\mathrm{Def}(H)=\sum_{e\in E(H)}\delta_H(e).$$

Double counting reciprocal incidences gives

$$|V^+(H)|=\frac{r}{D}|E(H)|+\mathrm{Def}(H), \tag{7}$$

where $V^+(H)$ is the set of nonisolated vertices.

Corollary 10.1 (Growing-crown Turán gap)**.** *There are absolute constants $Q_0<\infty$ and $c_0>0$ such that*

$$q \geq Q_0t^2 \quad\Longrightarrow\quad c^{\mathrm{lin}}_{q+1,t+1} \leq t-\frac{c_0}{t},$$

where

$$c^{\mathrm{lin}}_{r,k}=\limsup_{n\to\infty}\frac{\mathrm{ex}^{\mathrm{lin}}_r(n,C^r_{1,k})}{n}.$$

Proof. Take Q_0,ε_0 from Corollary 9.2, and put

$$A=\frac{\varepsilon_0}{t^3},\qquad \lambda=\frac{1}{2},\qquad b=\frac{1}{100t^2}.$$

Let

$$\mathcal{G}=\{e\in E(H):\delta_H(e)<A\},\qquad m=|E(H)|,\qquad x=|\mathcal{G}|.$$

Edges outside $\mathcal{G}$ give

$$\mathrm{Def}(H)\geq A(m-x).$$

Every $e\in\mathcal{G}$ has at least $\lambda r(D-1)$ neighboring witness edges of defect at least b. A fixed witness f is counted for at most $r(D-1)$ members of $\mathcal{G}$. Indeed, if $e\in\mathcal{G}$ meets f at v, degree stability at e gives $d_H(v)\leq D$; for each of the r points of f there are at most $D-1$ other edges through that point. Linearity counts an edge at most once, because two distinct points of f cannot lie on the same other edge. There are therefore at least λx distinct witnesses, and

$$\mathrm{Def}(H) \geq \lambda b x.$$

Consequently

$$\mathrm{Def}(H) \geq \max\{A(m-x), \lambda b x\} \geq \frac{A\lambda b}{A+\lambda b}\, m.$$

Because $\varepsilon_0 \leq 1/100$ and $t \geq 2$,

$$\frac{A\lambda b}{A+\lambda b} \geq \frac{\varepsilon_0}{2t^3}.$$

Equation (7) now gives

$$\frac{|E(H)|}{|V(H)|} \leq \frac{1}{r/D + \varepsilon_0/(2t^3)}.$$

Write $y = D/r$. Under $q \geq Q_0 t^2$, one has $t/2 \leq y \leq t$. Hence

$$\frac{1}{r/D + \varepsilon_0/(2t^3)} = \frac{y}{1 + y\varepsilon_0/(2t^3)} \leq \frac{t}{1 + \varepsilon_0/(4t^2)} \leq t - \frac{\varepsilon_0}{8t},$$

using $\varepsilon_0 \leq 1$. The result follows with $c_0 = \varepsilon_0/8$. □

11 Conclusion

Theorem 9.1 shows that the equality configuration is locally repelling when $t^2/q \to 0$: if one edge has $t^3\delta_H(e) = o(1)$, then almost all of its neighbors have defect at least a fixed multiple of t^{-2}. Corollary 10.1 turns the uniform form of this statement into a deficit of order $1/t$ from the coefficient t.

The two error scales in the local theorem have distinct sources. The term t^2/q measures the loss caused by the t colors used in a blocking matching, while $t^3\delta_H(e)$ controls the accumulated degree shortfall on the base edge. Extending the argument beyond $t^2/q = o(1)$ would require a sharper use of the pair coverage contributed by the colors in the blocking matching.

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